CISC839 Assignment 1: Analyzing Review Effort in Human vs. Agentic Pull Requests

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Part 1: Paper Review

Selected Paper: *Early-Stage Prediction of Review Effort in AI-Generated Pull Requests*

1. **Main Analytical Questions:**
   The paper investigates two core questions: (1) whether review effort for AI-generated pull requests can be predicted at creation time using only zero-latency structural features, and (2) which characteristics of agent-generated contributions lead to *agentic ghosting*, defined as PR abandonment after human feedback.
   Under the **SMART** framework, both questions are well-structured and practically relevant. The first is particularly strong due to its strict T_0 constraint, ensuring that only features available at submission are used, thereby eliminating data leakage and enabling real-time triage deployment. This design aligns well with real-world software engineering workflows, where early-stage decision support is critical. The second introduces a more rigorous definition of ghosting by combining rejection, explicit human feedback, and a 14-day inactivity window, improving upon simplistic merge-based proxies and capturing genuine interaction breakdowns.
   Despite this, both questions rely on a weak conceptualization of review effort. By defining effort as aggregated interaction volume (comments, reviews, and automated outputs), the study conflates system-generated noise with actual human cognitive work. This leads the model to prioritize high-activity PRs rather than truly complex ones, overestimating noisy cases (e.g., CI-triggered activity) and underestimating subtle but cognitively demanding changes. A more valid formulation would isolate human-authored, high-signal interactions within a controlled temporal window, thereby aligning the metric more closely with the intended construct of cognitive effort.
2. **Data Preparation:**
   The analysis is based on the AIDev dataset, comprising **33,596** agent-authored PRs from **2,807** repositories and **6,618** human-authored PRs for benchmarking. To reduce survivorship bias and ensure project maturity, the dataset is restricted to repositories with ≥**100** GitHub stars, which increases the likelihood of consistent review practices.
   A critical methodological step is the distinction between generative AI agents and deterministic maintenance bots. The authors apply heuristic filtering using metadata and naming patterns (e.g., Codex, Copilot, Devin, Claude), while excluding tools like Dependabot. This separation is essential: maintenance bots produce predictable, low-cognitive changes, whereas AI agents generate complex contributions that significantly impact review effort. Any conflation between these groups would severely bias the results and undermine the study's validity.
   Feature engineering includes **35** features across **complexity**, **intent**, and **context**. The temporal separation into T_0 (creation-time) and T_1 (pre-review) is a key strength, ensuring that no post-review information leaks into the predictive model. This design reflects strong methodological discipline and directly supports deployment in real-time triage systems.
   The target variable labels the top **20%** of PRs by total interaction volume as high-effort. While this provides a scalable labeling strategy, it introduces measurement noise, as interaction count is only a proxy and may not faithfully represent the true cognitive burden experienced by reviewers.
3. **Two Most Critical Limitations:**
   Limitation 1: Construct Validity of the Effort Metric:
   The primary limitation lies in how review effort is operationalized. By aggregating all interactions—including bot messages, CI/CD outputs, and formatting alerts—the Effort Score conflates automated activity with human cognitive effort. This constitutes a direct construct validity issue, as the metric captures volume rather than meaning.
   As a result, the model learns to predict structural noise instead of genuine review difficulty. PRs with heavy automated activity are overestimated, while logically complex but low-interaction changes are underestimated. Moreover, the study's sensitivity analyses do not resolve this issue, as they operate within the same flawed definition. A proper solution would require separating human-authored review signals from automated outputs before constructing the target variable, thereby restoring alignment between measurement and intent.
   Limitation 2: Structural Bias and Size Tautology:
   The model relies heavily on volumetric features (e.g., lines changed, files modified) while excluding semantic analysis due to latency constraints. This creates a circular dependency where size predicts effort because interaction volume itself scales with size, often amplified by CI/CD processes. This "size tautology" leads to systematic misclassification: large but trivial changes are flagged as high-effort, while small yet critical modifications—such as subtle security vulnerabilities—are underestimated. This semantic blind spot undermines the model's effectiveness in prioritizing true review risk and limits its practical reliability in real-world. Addressing this limitation requires incorporating lightweight semantic proxies (e.g., cyclomatic complexity or entropy-based measures) that can be computed within T_0 constraints, enabling better alignment between structural signals and actual cognitive complexity without violating latency requirements.

Part 2: Review Effort Analysis

Task 1: Refine Research Question

- Refined Research Question: Within highly active open-source GitHub repositories (≥ 100 stars), does the review effort—measured comparatively by Time-to-Resolution and Merge Rate—differ significantly between agentic and human-authored pull requests? Furthermore, what are the underlying drivers of cognitive friction within Agentic PRs when evaluated through advanced proxy metrics such as Cognitive Dispersion and Rework Ratios?

- Defining the review effort metric(s) & Explanations:
Due to the structural constraints of the AIDev dataset—where detailed review artifacts (comments, reviews, commit diffs) are exclusively available for the curated Agentic PR subset and largely absent for the human control group—I utilize a two-tiered metric system:

- Primary Comparative Metrics: (TTR) & Merge Rate

- **Definition Time-to-Resolution (TTR):** I define TTR as the elapsed time in hours between the creation of a PR (`created_at`) and its final terminal state, whether (`merged_at`) or (`closed_at`).
- **Definition (Merge Rate):** A binary outcome indicating whether a PR was successfully integrated (`merged_at` is not null) or outright rejected.
- **Justification:** TTR and Merge Rate are the only robust, shared metrics available across both human and agentic core metadata tables in the dataset. Measuring speed (TTR) in isolation is dangerous; a short TTR could represent either a flawless contribution or a rapid rejection of flawed code. Coupling TTR with the Merge Rate serves as a comprehensive, universal proxy for review effort and success.

- Secondary Diagnostic Metrics (Agentic Deep-Dive):
To analyze the specific nature of the review effort strictly within Agentic PRs where enriched data is available, I define two novel metrics:

- **Cognitive Dispersion Index (CDI):** Calculated by multiplying the number of files modified by the complexity of the patch, derived from the (`pr_commit_details`) table.
- **Justification:** AI agents occasionally execute scattered or unnecessary configuration changes. Tracking files touched evaluates the mental burden placed on reviewers who must trace fragmented logic to ensure no unintended side effects occur.
- **Rework-to-Initial Churn Ratio (RICR):** The volume of code churn (additions + deletions via `pr_commits`) occurring after the initial review timestamp in the (`pr_timeline`), divided by the initial PR size.
- **Justification:** A high ratio indicates poor initial code quality, quantifying the severe "ping-pong effect" where the human reviewer expends significant effort directing the agent to refactor its initial submission.

- Methodological Constraints & Discarded Metrics:
A composite **Review Effort Index (REI)** was designed to measure cognitive friction by standardizing metrics via Z-scores:

$$REI = z(RIB) + z(RB) + z(RTD) + z(TRE) + z(RCL)$$

where RIB (Rework Intervention Burden), RB (Rejection Burden), RTD (Review Thread Depth), TRE (Time Resolution Effort), and RCL (Review Comment Load). Additional metrics included **Discussion Depth Index (DDI)** and **Verdict Resistance (VR)**.

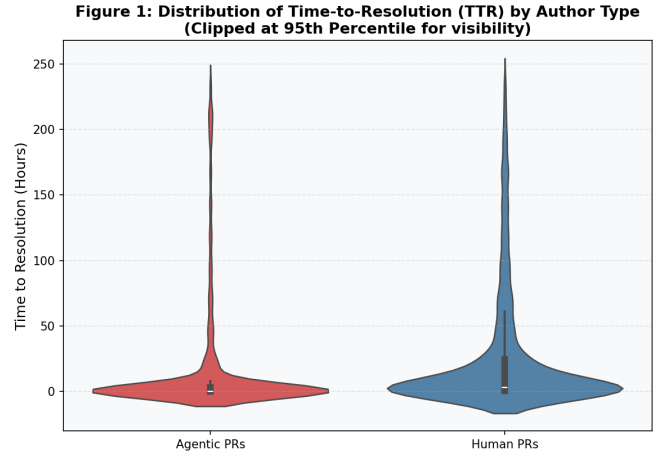
However, database limitations showed review data (`pr_reviews`, `pr_comments`) existed only for agentic PRs. Applying these metrics to human baselines would bias results (defaulting to zero). Thus, they were discarded in favor of TTR and Merge Rate, while still informing secondary metrics (CDI, RICR).

- Scope of Pull Requests to Analyze
To ensure empirical rigor, I have strictly defined the analysis scope based on the dataset's architectural boundaries:

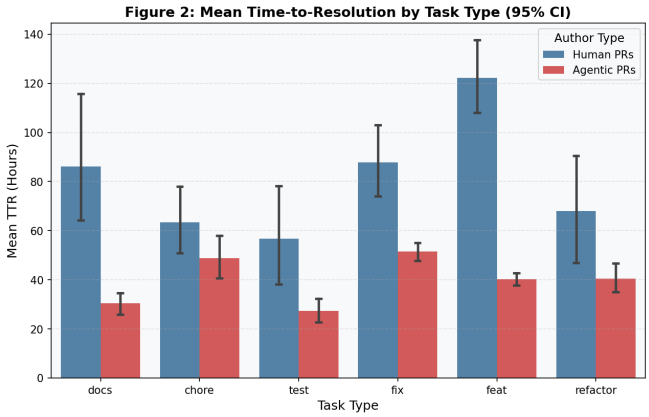
- **Repositories:** The analysis is strictly confined to the curated subset of 2,807 repositories possessing strictly > 100 GitHub stars. This ensures the analysis focuses on professional environments with established review cultures.
- **Time Period:** I include all pull requests available in the dataset up to the established cut-off date of August 1, 2025.
- **Selection Criteria (Agentic PRs):** I analyze the 33,596 Agentic-PRs authored by the five recognized AI coding agents: OpenAI Codex, Devin, GitHub Copilot, Cursor, and Claude Code.
- **Selection Criteria (Human PRs):** I utilize the exact baseline sample provided in the (`human_pull_request`) dataset (6,618 rows), ensuring perfect alignment with the > 100 stars repository ecosystem without introducing external sampling bias.

Task 2: Exploratory Data Analysis

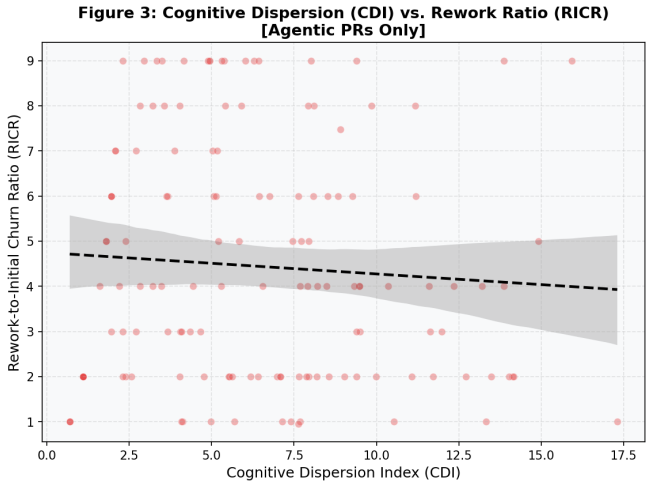
Four visualizations characterize the distributional properties of the primary metrics and probe the structural mechanisms driving review friction within the agentic subset.



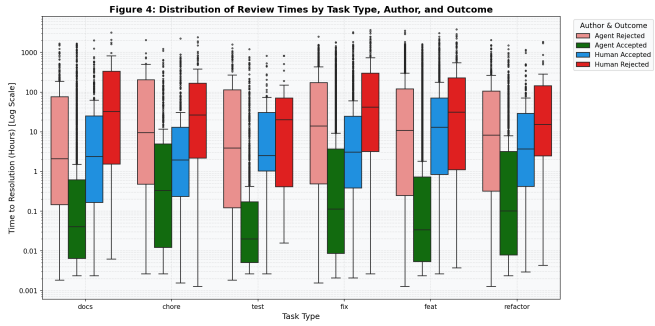
Agentic PRs resolve quickly on average, with occasional extreme delays; human PRs show slower, more balanced reviews.



Agentic PRs have lower mean TTR across all tasks. Docs resolve fastest, while fix and refactor take longer, reflecting task complexity.



A positive correlation is observable between file-level scatter (CDI) and post-review code churn (RICR). Scattered AI changes lead to successive correction cycles, capturing what we term *architectural diffusion*, which imposes compounding cognitive costs on reviewers.



Log-scale TTR across six task types by author and outcome. Accepted agentic PRs resolve fastest, rejected PRs show higher variance. `fix` and `refactor` have the widest IQR, reflecting higher complexity.

Table 1 summarizes the primary descriptive statistics for both cohorts.

Table 1: Primary Review Metrics by Author Type

Metric	Agentic PRs	Human PRs
Total PRs	31,284	6,149
Merge Rate	76.8%	82.6%
Mean TTR (hrs)	40.43	92.97
Median TTR (hrs)	0.12	5.36
Std. Dev. (hrs)	145.61	284.57
95th Pct. (hrs)	214.85	466.82

A critical paradox emerges: agentic PRs resolve faster on average yet merge at a lower rate, warranting formal hypothesis testing to disentangle speed from quality.

Task 3: Hypothesis Testing

H_0: $\mu_{\text{Agentic}} \leq \mu_{\text{Human}}$ $\quad H_A$: $\mu_{\text{Agentic}} > \mu_{\text{Human}}$

Test: Welch's one-tailed t-test, selected for robustness to unequal sample sizes ($N_{\text{Ag}} = 31{,}284$; $N_{\text{H}} = 6{,}149$) and variances ($\sigma^2_{\text{Ag}} \approx 21{,}202$; $\sigma^2_{\text{H}} \approx 81{,}060$). Unlike Student's t-test, Welch's formulation does not assume variance homogeneity, making it the statistically conservative choice for this heteroscedastic setting.

Table 2: Welch's t-Test Results (One-Tailed)

Quantity	Value
Mean TTR — Agentic (hrs)	40.43
Mean TTR — Human (hrs)	92.97
Test Statistic (t)	−14.12
p-value (one-tailed)	1.000

Conclusion: The test *fails to reject* H_0 (p=1.000, α=0.05). Agentic PRs resolve significantly faster (mean TTR ≈54 hrs lower), yet this does not imply lower review burden: the lower Merge Rate (76.8% vs. 82.6%) indicates many fast resolutions reflect rapid *rejection*— consistent with *reverse automation bias*, where reviewers quickly discount AI output and close PRs with minimal deliberation. *In this sense, I argue that speed reflects not efficiency of understanding, but a collapse of deliberation; the effort is not eliminated, but redistributed—from interpreting intent to rapidly validating or dismissing uncertain artifacts.*

Task 4: Regression Analysis

OLS regression on log(1 + TTR) controls for task-type confounders with HC3 robust standard errors:

$$\log(1 + \text{TTR}_i) = \beta_0 + \beta_1 \cdot \text{IsAgentic}_i + \sum_k \gamma_k \cdot \text{TaskType}_{ik} + \varepsilon_i$$

`chore` serves as the reference category; log-transformation addresses the severe right skew in the raw TTR distribution.

Table 3: OLS Results — DV: $\log(1 + \text{TTR})$, HC3 S.E.

Predictor	Coeff (β)	Std. Error	Z	p-value
Intercept	2.335	0.051	45.45	< 0.001
`is_agentic`	−1.058	0.032	−33.06	< 0.001
`task_docs`	−0.334	0.057	−5.83	< 0.001
`task_feat`	−0.037	0.053	−0.70	0.483
`task_fix`	0.203	0.054	3.74	< 0.001
`task_refactor`	0.085	0.063	1.34	0.180
`task_test`	−0.448	0.061	−7.37	< 0.001

$N = 34{,}992$; $R^2 = 0.051$; $F(6, 34985) = 296.5$, $p < 0.001$. Std. errors are HC3 robust.

Interpretation: $\hat{\beta}_{\text{IsAgentic}} = -1.058$ ($p < 0.001$) shows agentic PRs resolve $\sim 2.9\times$ faster than human PRs across task types. Bug fixes take longer, documentation and tests resolve faster, and feature/refactor effects are not significant.

Critical Limitation of Fit: The exceedingly low $R^2 = 0.051$ dictates that this model explains merely 5% of the variance in review time, leaving 95% driven by unobserved factors. I must critically note that relying solely on PR task type and AI authorship is entirely insufficient for practical, real-world TTR prediction. Furthermore, while the coefficients are statistically significant ($p < 0.001$), this is largely an artifact of the massive sample size ($N = 34,992$); the practical effect size remains structurally weak despite the statistical certainty.

Task 5: Summary of Findings and Limitations

Interpretive Insight: Agentic PRs transform rather than reduce review effort. Reviewers shift from interpreting human intent to interrogating opaque artifacts, replacing understanding with calibrated trust. Faster resolutions coupled with lower acceptance rates indicate that cognitive effort is redistributed into rapid, cautious triage rather than eliminated.

Key Findings ($N = 37,433$ PRs, ≥100 star repos):

(1) **Speed–quality paradox:** Agentic TTR (40.4 hrs) halves human TTR (93.0 hrs), but merge rates are lower (76.8% vs. 82.6%), indicating rapid TTR often reflects swift rejection rather than frictionless approval.

(2) **Authorship accelerates TTR:** $\hat{\beta}_{\text{IsAgentic}} = -1.058$ ($p < 0.001$) holds across tasks, suggesting either AI efficiency or a reverse *automation bias* where reviewers rapidly dismiss AI output.

(3) **Diffusion drives rework:** CDI–RICR analysis proves that file-scattered contributions necessitate post-review corrections. Structural incoherence—not raw PR size—is the primary driver of AI review friction.

Limitations:

(1) **Dataset asymmetry:** Missing human review artifacts forces reliance on TTR, which conflates active review effort with passive queue time, potentially inflating human TTR estimates.

(2) **Organizational confounding:** Repositories with mature AI cultures may utilize automated pre-screening, artificially compressing agentic TTR independent of actual code quality.

Part 3: GenAI-Augmented Reanalysis

Task 1: AI Usage Explanation

The GenAI pipeline used **Google Gemini 2.5 Pro** to rebuild the regression component (Part 2, Task 4), acting as a co-engineer through two targeted prompts.

P1 — Weakness Diagnosis: *My OLS predicts log(1+TTR) with is_agentic + task_type dummies, R^2=0.051. What are the weaknesses? Suggest 3 improvements."* Gemini identified: (1) TTR conflates active review with queue time; (2) low R^2 signals omitted confounders; (3) binary merge outcome captures review quality more directly than duration. This motivated switching to Logistic Regression on `is_merged`.

P2 — Model Build: *Write statsmodels Logit with HC3 robust SE predicting is_merged using is_agentic and task_type dummies. Return odds ratios and 95% CIs."* Gemini returned executable code run directly in the notebook, converging in 6 iterations and yielding the results below.

Task 2: Required Comparison

Pipeline. The baseline measures *how long* review takes via OLS on log(1+TTR); the GenAI pipeline measures *whether it succeeds* via Logit on `is_merged`—a shift from duration-based to outcome-based operationalization of review effort.

Results. OLS shows agentic PRs resolve 2.88× faster ($\hat{\beta}$=−1.058, $p < 0.001$), suggesting lower burden in isolation. Logit recontextualizes this: agentic PRs have 40.8% lower merge odds (0.592, $p < 0.001$), revealing a *fast-rejection dynamic* where speed reflects discard rather than quality. Bug-fix PRs prolong TTR in OLS but are not disproportionately rejected in Logit (1.052, $p = 0.424$). Documentation PRs reduce TTR *and* increase merge odds (2.589, $p < 0.001$), confirming low architectural risk. Fit: OLS $R^2 = 0.051$; Logit Pseudo-$R^2 = 0.014$; $n = 34,992$.

Improvement? Yes, partially. Logit uncovers the fast-rejection dynamic invisible to OLS, providing genuine diagnostic value. Yet the low Pseudo-R^2 shows that authorship and task type alone explain little variance in merge success; organizational confounders remain unmodelled in both pipelines.

Bonus: Analytical Mistake Introduced by AI

When prompted to enrich the model with `pr_reviews` and `pr_comments` counts, Gemini merged these tables into `master_df` and applied `fillna(0)` to human rows—unaware that these tables contain *only agentic PR data*. Human NaNs represent missing data, not zero activity; the model incorrectly treats them as evidence of no review work, contradicting empirical reality (human mean TTR = 92.97 hrs).

The fallout is twofold: (1) **near-perfect multicollinearity**—`comment_count` ≡ 0 whenever `is_agentic`= 0, so OLS cannot disentangle authorship from comment effects; (2) **double-encoding of author type**—via `is_agentic` and the zero-imputed proxies simultaneously—spuriously amplifying the human penalty coefficient. Gemini produced syntactically valid code that unintentionally violates construct validity, undetectable by automated tests.